SECRET AGENT

Case File

Words only. No pictures. Build it right.



AGENT ID

Name: _____

Partner: _____

Event: Describe It, Build It

If found, please return to the DDMES Science Olympiad binder shelf.

How a build agent works

In Describe It, Build It, one agent (the **Describer**) studies a mystery object and writes secret directions. The other agent (the **Builder**) is in a different room and rebuilds it from the words alone. No pictures allowed! This case file lives in your SciOly binder.

1
STUDY
Look at the object from every side. Pick the front.

2
NAME
Give every piece a full name: color, size, type.

3
PLACE
Say exactly where each piece goes.

4
CHECK
Add check sentences so your Builder can catch mistakes.

5
BUILD
Builder: read it all, sort the parts, build one step at a time.

The Agent's Code

- [ ] Words and numbers only. No drawings, arrows, codes, or letter shapes.
- [ ] Define every abbreviation first: rt = right.
- [ ] The Describer stays silent while the Builder builds. No hints, no pointing!
- [ ] Builders never peek at the model.
- [ ] Score honestly: a point only if the piece is in the right place AND connected right.
- [ ] Switch roles so both partners practice.

My agent kit

- [ ] This case file and 2 sharp pencils
- [ ] Colored pencils: black, brown, green, blue, red
- [ ] LEGO bricks or craft pieces (2 matching sets)
- [ ] A ruler with centimeters
- [ ] A timer
- [ ] Wide-ruled paper

Journal map

Know your event	p. 3-6
Cases #1 to #4	p. 7-10
Template, log, judging	p. 11-13
Tournament day	p. 14
Missions and badges	p. 15-16
Answer key (upside down)	p. 17

Where to practice

The kitchen
cups, spoons, boxes, and cans to stack and describe

The toy box
LEGO, K'nex, Tinker Toys, or wooden blocks

The craft drawer
straws, pipe cleaners, craft sticks, tape

The recycling bin
boxes, lids, and tubes for a surprise build

A LEGO booklet
cover the pictures and build from words only

Your family
perfect Builders who have never seen the model!

Did you know? At the tournament, breaking a writing rule (like drawing one arrow) drops your team below EVERY team that followed the rules, even if the build is perfect.

Mission briefing: the event

The Describer gets 25 minutes to write build directions for a mystery object using only words and numbers (3 sheets of wide-ruled paper); then the Builder, in a different room, has 20 minutes to rebuild it from those words, and the closest match wins.

The Describer

- Sees the mystery object.
- Has **25 minutes** to write.
- Gets 3 sheets of wide-ruled paper (front and back).
- Words and numbers only.

The Builder

- Waits in a different room.
- Has **20 minutes** to build from the words.
- May bring colored pencils: black, brown, green, blue, red.
- Build time breaks ties.

The rules in one box

Describer	Sees the object and has 25 minutes to write directions. Gets 3 sheets of paper (front and back).
Builder	In a different room. Has 20 minutes to build from the directions. May use colored pencils (black, brown, green, blue, red).
Only words and numbers	No drawings, diagrams, arrows, or made-up symbols. Normal punctuation is fine.
Abbreviations	Allowed if you define them first: rt = right.
Scoring	Points for each piece in the right place and connected right. Breaking the writing rules drops you below every team that followed them.

How the judge scores

- [ ] **1 point** for each piece in the right place AND connected the right way.
- [ ] A piece still counts if an earlier piece was wrong.
- [ ] Extra pieces cost nothing. A missing piece just gets 0.
- [ ] Same score? The **faster build** wins.
- [ ] Broke a writing rule? You rank below every team that followed the rules.

Tournament time plans

Describer: 25 minutes	Builder: 20 minutes
2 min: study the object from every side, set the front	2 min: read the whole description
3 min: abbreviations, overview, parts list	2 min: sort parts and match the parts list
15 min: numbered steps, bottom up	14 min: build one step at a time, check off each step
5 min: reread and add check sentences	2 min: compare to every check sentence

Heads up: the mystery object might be LEGO, or K'nex, Tinker Toys, Lincoln Logs, or craft materials like straws, cups, and pipe cleaners. Practice with all of them!

Legal or not legal?

Allowed

- Words and numbers
- Normal punctuation: . , ? ( ) :
- Abbreviations you define first (rt = right)
- Numbers for size and count: 2 by 4, 3 cm

Not allowed

- Drawings, diagrams, arrows (->)
- Letters or numbers used as pictures (bend it into a V)
- Made-up symbols and codes (@, :) faces)
- An x for “by”: write 2 by 4, not 2x4

Test yourself

Circle one for each. Then check the answer key on p. 17. Get all 7 right for the **Rule Keeper** badge.

1	Put the red brick -> on the blue brick.	◯ Legal ◯ Not legal
2	rt = right. Put the cup on the rt side of the stick.	◯ Legal ◯ Not legal
3	Stack 2 yellow bricks on the green plate.	◯ Legal ◯ Not legal
4	Bend the pipe cleaner into a V.	◯ Legal ◯ Not legal
5	Use the blue brick (the one with 8 bumps).	◯ Legal ◯ Not legal
6	Tape the straw @ the back of the cup.	◯ Legal ◯ Not legal
7	Put 2 stickers on the cup like a smiley face :)	◯ Legal ◯ Not legal

Fix it!

Rewrite each sentence so it follows the rules and a Builder would know exactly what to do.

1. “Put the blue brick on the red brick like a T.”

2. “Stand the straw ^ in the cup.”

3. “Put the cup over there, next to the other thing.”

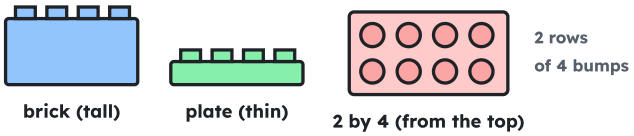
4. “Bend the pipe cleaner into a U.”

Agent word bank

Part name recipe

Color	red, dark blue, light green (the judge will say if color does not matter)
Size	2 by 4 (2 rows of 4 bumps), 10 centimeters long
Type	brick (tall), plate (thin), straw, craft stick, cup
Special features	bendy end, one flat side, a hole in the middle

Example: **a red 2 by 4 LEGO brick**, or **a bendy straw, 21 centimeters long**.



Count the bumps on top. Write sizes with “by”: 2 by 4.

Position words

Check each word after you use it in real directions.

☐ on top of ☐ underneath ☐ left end ☐ right end ☐ front ☐ back ☐ centered ☐ lined up with
☐ hanging over by 1 bump ☐ lying flat ☐ standing up ☐ upside down ☐ parallel (side by side, same direction)
☐ perpendicular (crossing to make square corners) ☐ touching ☐ between ☐ corner ☐ edge

Craft words

Straw	long or short part, bendy part, open end
Pipe cleaner	wrap (go around), twist (turn two ends together), bend in half, bend to make a square corner
Cup	rim (top edge), bottom, upright or upside down, inside, outside
Craft stick	flat side, thin edge, rounded end
Measuring	centimeters from the end, halfway, one third of the way, touching, 2 finger widths is NOT exact

Set the front first! “Lay the blue brick flat with its long side facing you. That long side is the front.” Now left, right, front, and back mean the same thing to both agents.

The 6-part plan

Every strong description has the same 6 parts in the same order, so your Builder always knows what comes next.

1
Abbreviations: define them at the very top (br = brick).

2
Overview: what it looks like overall, how big, how many pieces.

3
Parts list: every piece with its full name and how many.

4
Front: tell the Builder how to turn the base so everyone agrees on front, back, left, and right.

5
Steps: bottom up, one piece per step, numbered.

6
Checks: every few steps, a sentence the Builder can use to check their work.

Sample description

Abbreviations: br = brick. Overview: This object is 3 layers tall and made of 4 LEGO pieces. Parts: 1 blue 2 by 4 br, 2 red 2 by 2 br, 1 yellow 2 by 4 br. Front: Lay the blue br flat with its long side facing you. That long side is the front. Step 1: Put 1 red br on the left end of the blue br, covering its 4 left bumps. Step 2: Put the other red br on the right end the same way. Check: your object is now 2 layers tall. Step 3: Put the yellow br on top, lined up exactly with the blue br. Check: from the front you see blue on the bottom, 2 reds, then yellow on top.

Find all 6 parts in the sample. Underline every part name. Circle the 2 check sentences.

Good check sentences

Height	Now your object is 3 layers tall.
What you see	From the front you see red, then blue, then red.
Count	You should have used 6 pieces so far.
Leftovers	You should have 2 craft sticks left.

A good check says what the Builder should **see**, not just “check your work.”

Try it: write an overview

Pick something on your table (a pencil box, a mug, a stack of books). Write its overview: what it looks like, how big, how many pieces.

Case #1: The Kitchen Stack

TOP SECRET

ASSIGNED after Lesson 1 (Fri Oct 16) · **BADGE** **Kitchen Describer**

YOUR MISSION Stack or line up 5 things from your kitchen (spoon, cup, box, and so on). In 10 minutes, write directions to rebuild it using only words and numbers. Take it apart and have a family member rebuild it from your words. Write down one word that confused them.

1. My 5 kitchen pieces (full names)

1. _____
2. _____
3. _____
4. _____
5. _____

Secret sketch of my stack (Builders, no peeking!)

2. My directions (10 minutes, words and numbers only)

3. Score the rebuild

Piece	Right place?	Connected right?	Point?
Total			/ 5

The word that confused my Builder: _____

A clearer way to say it: _____

Case #2: Name It and Place It

TOP SECRET

ASSIGNED after Lesson 2 (Fri Nov 20) · **BADGE** **Part Namer**

YOUR MISSION Pick 3 things at home and say a full name for each out loud (color, size, type, special features). Then play Hidden Objects with a family member.

Part A: Full names

Write the recipe for each thing. Example: *white, 10 cm tall, paper cup, has a rim.*

Thing	Color	Size	Type	Special features

Part B: Hidden Objects

Put 4 small things (a spoon, a cup, a toy, a sock) on a table in a secret arrangement. Your partner waits in another room. Write where each thing goes using only **position words**. Your partner sets them up from your words.

_____

_____

_____

_____

_____

_____

_____

_____

Position words I used

- ☐ on top of
- ☐ underneath
- ☐ left end
- ☐ right end
- ☐ centered
- ☐ lined up with
- ☐ touching
- ☐ between
- ☐ parallel
- ☐ perpendicular
- ☐ upside down
- ☐ standing up

How many things were in the right place?

___ / 4

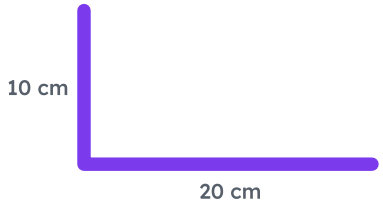
Case #3: The Pipe Cleaner Caper

TOP SECRET

ASSIGNED after Lesson 3 (Fri Jan 15) · **BADGE** **Craft Measurer**

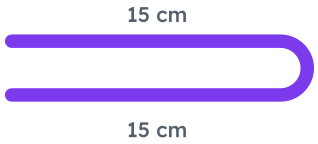
YOUR MISSION Craft objects need exact words and centimeters. Describe each bent pipe cleaner using only words and numbers. No letter shapes (not “an L”!). Then make your own.

Shape A



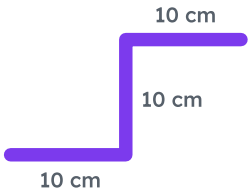
Words-only directions:

Shape B



Words-only directions:

Shape C



Words-only directions:

Every pipe cleaner is 30 cm long. Compare with the sample answers on p. 17 (different words are fine if they are clear and follow the rules).

Your turn: bend, describe, rebuild

Bend a pipe cleaner (or a strip of paper) into a secret shape with 2 or 3 bends. Write directions with centimeters. A family member bends a new one from your words. Measure how far off each bend is.

My directions:

Secret sketch of my shape (only I see it)

Bend	How far off (cm)
1	
2	
3	

1 cm or less off = a strong description!

Case #4: The Final Test

TOP SECRET

ASSIGNED after Lesson 4 (Fri Feb 12) · **BADGE** **Check Writer**

YOUR MISSION Do one last practice at home: Describers write for 15 minutes about an object made of 8 to 10 pieces; Builders rebuild from the written directions. Put your final checklist in the front of this divider to read the morning of the tournament.

Date _____ **My role** _____ **Pieces** _____

Describers: my 3 check sentences

1. **Height:** _____
2. **From the front:** _____
3. **Count:** _____

Builders: my build

Build time: _____ Did I read everything first? _____

Pieces right: _____ out of _____ Did I check off each step? _____

Our results

Pieces correct	Pieces possible	Build time	Rule problems?

The first mistake happened at step ____ because:

_____

_____

Our plan for tournament day:

_____

_____

_____

Put your finished tournament checklist (p. 14) at the front of your binder divider to read the morning of the tournament.

Directions template

Use this page for a full practice description (Lab I1, Case #4, or any build). Fill every part to earn the **Plan Master** badge.

Object _____ **Pieces** _____ **Date** _____

1. Abbreviations

2. Overview (what it looks like, how big, how many pieces)

3. Parts list (full names and how many)

4. Front (how to turn the base)

5. Steps (bottom up, one piece per step)

1.
2.
3.
4.
5.
6.
7.
8.
9.
10.

6. Checks (height, what you see, a count, leftovers)

Evidence: Build Log

Log every practice build: at meetings, in labs, and at home. D = Describer, B = Builder. Score it like a judge. Log 3 builds that get faster for **Speed Builder**, score 3 builds for **Fair Judge**, and log both roles for **Role Switcher**.

Date	D or B	Object (LEGO or craft)	Pieces	Score	Build time	First mistake at step	What we'll fix

My best score so far: _____

My fastest build: _____

The Judge's Desk

CLASSIFIED

You are the event judge today. Read the model, then score each team out of 5 and rank them. Check the answer key on p. 17.

The model (5 pieces)

A green 2 by 4 plate lies flat, long side facing you (that is the front). A red 2 by 2 brick sits on the left end of the plate. A blue 2 by 2 brick sits on the right end. A yellow 2 by 4 brick lies on top of the red and blue bricks, lined up with them. A white 1 by 2 brick sits in the middle of the yellow brick, going left to right.

Team	What the Builder built	Score	Time
X	Everything is right.		9 min
Y	Everything is right, and they added an extra black 1 by 1 brick on the plate.		12 min
Z	The red and blue bricks are switched (red on the right, blue on the left). Everything else is right.		7 min
W	Everything is right, but the Describer drew an arrow in the directions.		6 min

Rank: 1st _____ 2nd _____ 3rd _____ 4th _____

Why did Team W rank where it did?

Judge's rubric: score a real build

Piece	Right place?	Connected right?	Point
Total and build time			

Tournament day: Sat Mar 13

Read this page the morning of the tournament. Check each box when you really did it.

Describer checklist

- [ ] Abbreviations are defined at the top or the first time I use them.
- [ ] Overview, parts list, and front are at the beginning.
- [ ] Steps are numbered, bottom up, one piece per step.
- [ ] I added check sentences.
- [ ] No drawings, arrows, codes, or letter shapes anywhere.
- [ ] I reread everything from the Builder's point of view.

Builder checklist

- [ ] I read the whole description first.
- [ ] I sorted my parts and matched the parts list.
- [ ] I built one step at a time and checked off each step.
- [ ] I compared my build to every check sentence.

Builders bring

- [ ] Colored pencils: black, brown, green, blue, red (no other colors)

Pack the night before

- [ ] 2 sharp pencils each and an eraser
- [ ] Builder's colored pencils
- [ ] A snack and a water bottle
- [ ] A good night's sleep!

Our time plan

Describer: 25 minutes	Builder: 20 minutes
2 min: study the object from every side, set the front	2 min: read the whole description
3 min: abbreviations, overview, parts list	2 min: sort parts and match the parts list
15 min: numbered steps, bottom up	14 min: build one step at a time, check off each step
5 min: reread and add check sentences	2 min: compare to every check sentence

Good luck note to my partner

Case Missions

One case for each Describe It, Build It meeting. When you finish, show a coach or your Event Captain for initials, then color in the badge.

When	Mission	Badge	Done	Initials
After Fri Oct 16 Lesson 1	Case #1: The Kitchen Stack (p. 7).	Kitchen Describer	☐	
After Fri Nov 20 Lesson 2	Case #2: full names and Hidden Objects (p. 8). Get all 7 Legal or not right (p. 4).	Part Namer / Rule Keeper	☐	
After Fri Jan 15 Lesson 3	Case #3: The Pipe Cleaner Caper (p. 9).	Craft Measurer	☐	
After Fri Feb 12 Lesson 4	Case #4: The Final Test with 3 check sentences (p. 10).	Check Writer	☐	
Anytime Bonus	Write a full description on the Directions template (p. 11).	Plan Master	☐	
Anytime Bonus	Judge's Desk (p. 13) and score 3 builds in the Build Log (p. 12).	Fair Judge	☐	
Anytime Bonus	Log 3 timed builds that get faster, in both roles (p. 12).	Speed Builder / Role Switcher	☐	

Tournament day: Sat Mar 13

- ☐ 2 sharp pencils (Builders: colored pencils too)
- ☐ Your tournament checklist (p. 14)
- ☐ A good night's sleep

- **Describers:** use all 25 minutes. Reread as the Builder.
- **Builders:** read everything first, then build.
- No arrows, no letter shapes, no 2x4!

CASE CLOSED

This agent earned all 9 badges and is now a **Master Describer**.

Agent _____ Coach or Event Captain _____ Date _____

My Badges

Color in each badge when you earn it. Earn all 9 to become a **Master Describer**! Your Team Badge Passport has the same badges.



Rule Keeper

All 7 Legal or not right

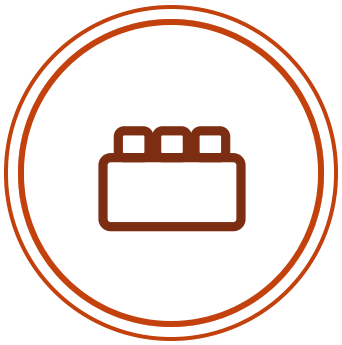
Initials: _____



Kitchen Describer

Case #1: The Kitchen Stack

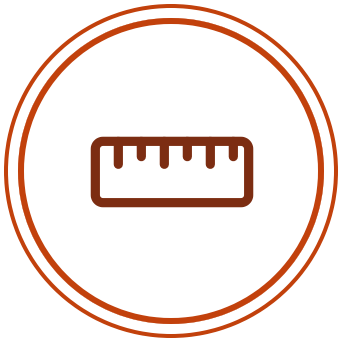
Initials: _____



Part Namer

Case #2: full names + hidden objects

Initials: _____



Craft Measurer

Case #3: pipe cleaners in cm

Initials: _____



Check Writer

Case #4 with 3 check sentences

Initials: _____



Plan Master

A full 6-part description

Initials: _____



Fair Judge

Judge's Desk + score 3 builds

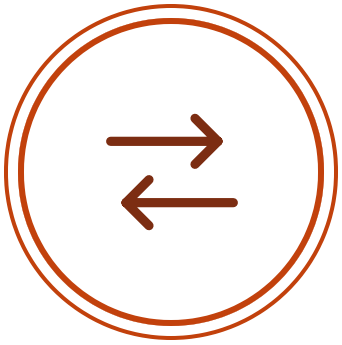
Initials: _____



Speed Builder

3 timed builds that get faster

Initials: _____



Role Switcher

Be the Describer AND the Builder

Initials: _____

Answer key

No peeking until you have tried the page!

Legal or not (p. 4)

1 Not legal (arrow symbol). 2 Legal (abbreviation defined first; this is the rules' own example). 3 Legal. 4 Not legal (a letter used as a picture). 5 Legal (parentheses used normally). 6 Not legal (@ used as a symbol). 7 Not legal (a face made from punctuation is a picture).

Fix it (p. 4), sample answers

1. “Lay the blue 2 by 4 brick on top of the red 2 by 4 brick, centered and perpendicular (crossing it to make square corners).” 2. “Stand the straw straight up inside the cup, touching the bottom.” 3. “Put the cup on the right side of the plate, touching the spoon.” (Name the real things and use position words.) 4. “Bend the pipe cleaner in half into a smooth curve, so the two ends point the same way, 3 centimeters apart.”

The Pipe Cleaner Caper (p. 9), sample answers

A: Lay the pipe cleaner flat. Bend it 10 centimeters from one end to make a square corner. **B:** Bend the pipe cleaner in half so the two halves lie side by side, parallel and close together. **C:** Lay it flat. Bend a square corner 10 centimeters from the left end so the middle part goes straight up. Bend another square corner 10 centimeters later so the last 10 centimeters point right, parallel to the first part (like one stair step).

Judge's Desk (p. 13)

X: 5 (9 min). Y: 5 (12 min; extra pieces cost nothing). Z: 3 (red and blue are in the wrong places). W: 5, but broke a writing rule.

Rank: 1st X (ties Y at 5 but built faster), 2nd Y, 3rd Z, 4th W. Team W drew an arrow, so it ranks below every team that followed the rules, even with a perfect build.

The 6-part plan (p. 6)

Part names in the sample: blue 2 by 4 brick, red 2 by 2 brick (two), yellow 2 by 4 brick. Checks: “your object is now 2 layers tall” and “from the front you see blue on the bottom, 2 reds, then yellow on top.”